

Load balancer task



AWS Load Balancing Project Documentation

This document contains all AWS tasks completed as part of the load balancing and networking project.

Each task is written in a **standard professional format** including:

- Task Title
- Objective
- Prerequisites
- Step-by-step implementation with evidence
- Validation steps
- Issues faced
- Conclusion

The purpose of this document is to create a clear and complete record of how different AWS load balancers and related services were configured.

Project Overview

In this project, we configure the following AWS components:

- Classic Load Balancer (CLB)
- Application Load Balancer (ALB)
- Network Load Balancer (NLB)
- SSL Certificate integration using ACM
- Route 53 domain mapping
- Load balancer logging to S3

These tasks demonstrate high availability, traffic distribution, secure HTTPS setup, and architecture best practices.



Base Infrastructure Requirements

Before starting any task, the following AWS setup must already exist:

- A VPC
- Public Subnet(s)
- EC2 instance(s) to act as backend servers
- Security Groups with required ports open
- Internet Gateway attached to VPC
- Route Table configured for internet access
- Key Pair for EC2 login

This ensures that all load balancers can route traffic correctly.

Task Implementation Begins Below

1) Task 1: Configure Classic Load Balancer

Task Title

Configure Classic Load Balancer (CLB) on AWS

Objective

To create and configure a **Classic Load Balancer** that distributes incoming traffic to EC2 instances hosted in a public subnet.

Prerequisites

- ✓ VPC created
- ✓ Public Subnet
- ✓ EC2 Instances running in the same VPC

- ✓ Security Group allowing port 80
- ✓ Key Pair
- ✓ Internet Gateway attached to VPC
- ✓ Route Table with 0.0.0.0/0 pointing to IGW

Step-by-Step Implementation (With Evidence)

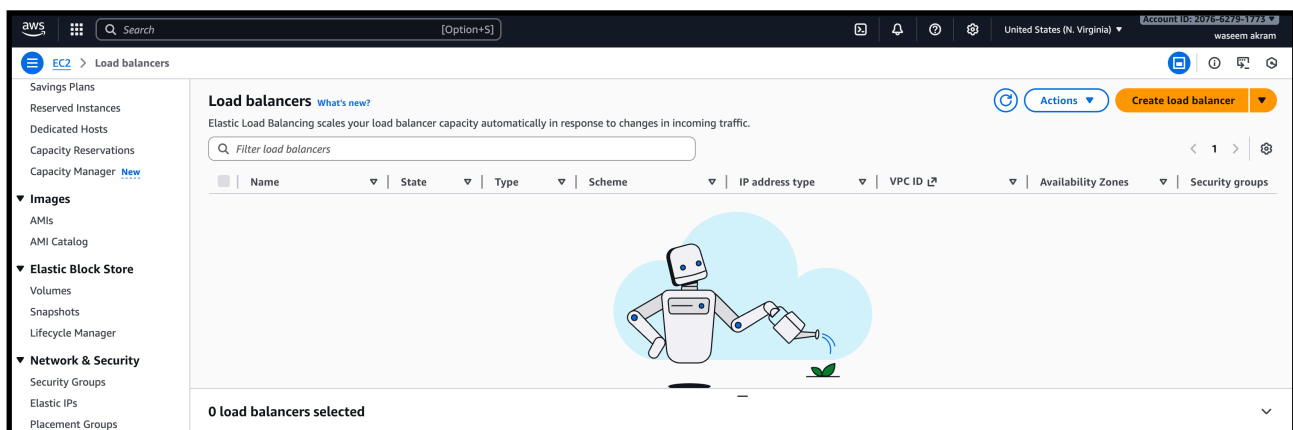
Step 1: Navigate to Load Balancers

Console Path:

EC2 Dashboard → Load Balancing → **Load Balancers** → *Create Load Balancer*

Explanation:

We start from the EC2 Load Balancer section to create a Classic Load Balancer.



Step 2: Select Classic Load Balancer

- Choose **Classic Load Balancer** option
- Click **Create**

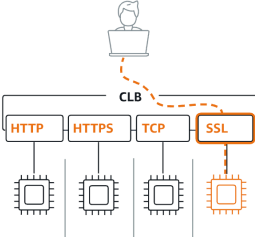
Create

latencies.
Create

▼ Classic Load Balancer - previous generation

Classic Load Balancer

Info



Choose a Classic Load Balancer when you have an existing application running in the EC2-Classical network.
Create

Close

CloudShell

Feedback

Console Mobile App

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Step 3: Configure Basic Settings

Fill the details:

Parameter	Value
Load Balancer Name	classic-lb
Scheme	Internet Facing
Listener	HTTP (80) → Instance Port 80

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

classic-lb

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme

Info

Scheme can't be changed after the load balancer is created.

Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.

Network mapping

Info

The load balancer routes traffic to targets in the selected subnets, and in accordance with your network settings.

VPC

Info

Select the virtual private cloud (VPC) for your targets. Only VPCs with an internet gateway are available for selection. The selected VPC cannot be changed after the load balancer is created. When selecting a VPC for your load balancer, ensure each subnet has a CIDR block with at least a /27 bitmask and at least 8 free IP addresses.

Learn more

vpc-07968786753c2ea72 (default)
172.31.0.0/16

(default)

Create VPC

Availability Zones and subnets

Select at least one Availability Zone and one subnet for each zone. We recommend selecting at least two Availability Zones. The load balancer will route traffic only to targets in the selected Availability Zones. Availability Zones that are not supported by the load balancer or the VPC are not available for selection.

us-east-1a (use1-az1)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-071d209e29a1a1f82
IPv4 subnet CIDR: 172.31.0.0/20

pubsubnet

IPv4 address

4: Assign Security Group

- Choose **existing security group** or create new one
- Allow **Inbound HTTP (80)**

Explanation:

—> This allows web traffic to reach the load balancer.

The screenshot displays the AWS Management Console interface for configuring a load balancer. It is divided into two main sections: 'Security groups' and 'Listeners and routing'.

Security groups: This section includes an 'Info' link and a description: 'A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).' Below this is a 'Security groups' dropdown menu with the text 'Select up to 5 security groups'. A selected group is shown in a box: 'default' with ID 'sg-0fbfec998b25506f' and VPC 'vpc-07968786753c2ea72'.

Listeners and routing: This section includes an 'Info' link and a description: 'A listener is a process that checks for connection requests using the protocol and port you configure. The settings you define for a listener determine how the load balancer routes requests to its registered targets.' It shows a single listener configuration for 'Listener HTTP:80' with 'Instance HTTP:80'. The configuration includes two columns of settings: 'Listener protocol' (HTTP) and 'Listener port' (80) on the left, and 'Instance protocol' (HTTP) and 'Instance port' (80) on the right. Below these is an 'Add listener' button and a note: 'You can add up to 99 more listeners.'

Step 5: Configure Health Checks

Use the following:

Field	Value
Ping Protocol	HTTP
Ping Port	80
Ping Path	/
Healthy Threshold	2
Unhealthy Threshold	2
Timeout	5 sec
Interval	10 sec

Explanation:

CLB will check if EC2 is healthy using the / path

Health checks [Info](#)

Your load balancer automatically performs health checks to test the availability of all registered instances. Traffic is only routed to healthy instances, which is determined on their response to the health check.

Ping target
The health check ping is sent using the protocol and port you specify. If using HTTP/HTTPS protocol, you must also provide the destination path.

Ping protocol

Ping path

Ping port

1-65535

► Advanced health check settings

Step 6: Register EC2 Instances

- Select your EC2 instances
- Click **Add to Load Balancer**

—> so lets launch two instances within the given vpc range subnets

—> below image is from first instance script and launch the instance with this script

EC2

Instances

Launch an instance

Metadata version [Info](#)

V2 only (token required)

⚠

For V2 requests, you must include a session token in all instance metadata requests. Applications or agents that use V1 for instance metadata access will break.

Metadata response hop limit [Info](#)

2

Allow tags in metadata [Info](#)

Select

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

⬆

Choose file

```
#!/bin/bash
sudo yum update -y
sudo yum install httpd -y
echo "hello from instance 1" >> /var/www/html/index.html
sudo systemctl start httpd
sudo systemctl enable httpd
```

☐ User data has already been base64 encoded

CloudShell

Feedback

Console Mobile App

—> below is from second instance

Applications or agents that use VTL for instance metadata access will break.

Metadata response hop limit | [Info](#)

2

Allow tags in metadata | [Info](#)

Select ▼

User data - optional | [Info](#)

Upload a file with your user data or enter it in the field.

[Choose file](#)

```
#!/bin/bash
sudo yum update -y
sudo yum install httpd -y
echo "hello from ec2 2 instance" >> /var/www/html/index.html
sudo systemctl start httpd
sudo systemctl enable httpd
```

—> after launching both instances we have to register this both instances inside load balancer

EC2 > Load balancers > Create Classic Load Balancer

Health checks [Info](#)

Your load balancer automatically performs health checks to test the availability of all registered instances. Traffic is only routed to healthy instances, which is determined on their response to the health check.

Ping target

The health check ping is sent using the protocol and port you specify. If using HTTP/HTTPS protocol, you must also provide the destination path.

Ping protocol HTTP **Ping path** /index.html

Add instances

Select EC2 instances to register to your load balancer. Requests will be routed to registered instances that meet the health check requirements. For maximum fault tolerance, we recommend maintaining approximately equivalent numbers of instances in each Availability Zone enabled for the load balancer. If demand on your instances changes, you can register or deregister instances without disrupting the flow of requests to your application. [Learn more](#)

VPC

vpc-07968786753c2ea72

Available instances (2/2)

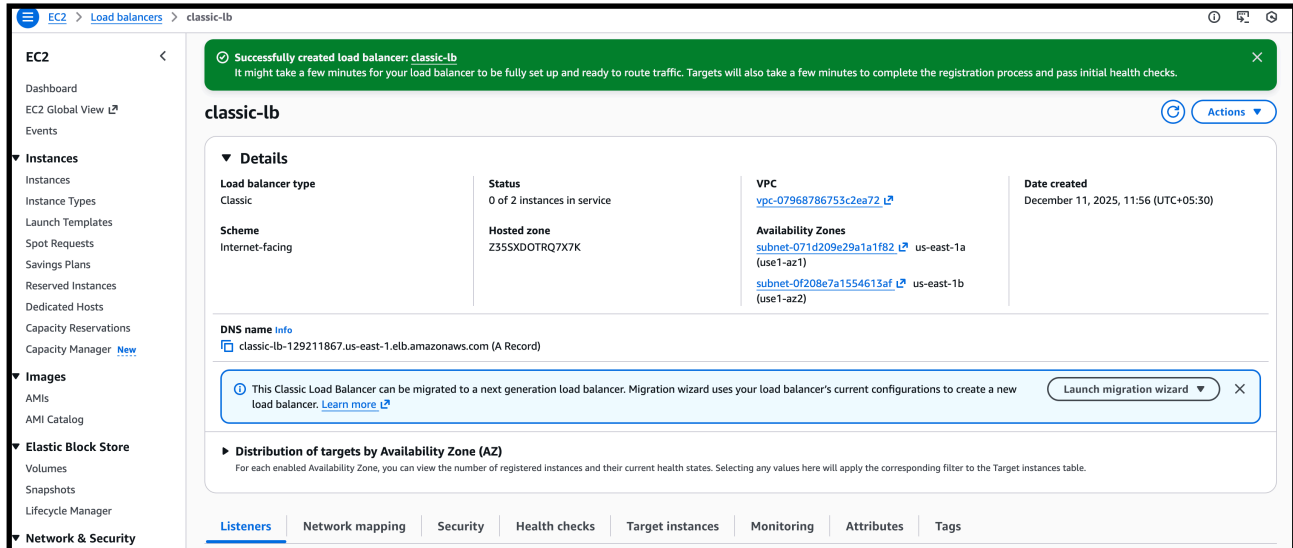
Filter available instances

☒	Instance ID	Name	State	Security groups	Zone	Public IPv4 address	Su
☒	i-08a86882c60d3a475	myec22-lb	Running	launch-wizard-6	us-east-1b	52.90.57.188	sut
☒	i-05c9df4b15d784e6	my-ec21-lb	Running	launch-wizard-5	us-east-1a	3.237.204.61	sut

[Cancel](#) [Confirm](#)

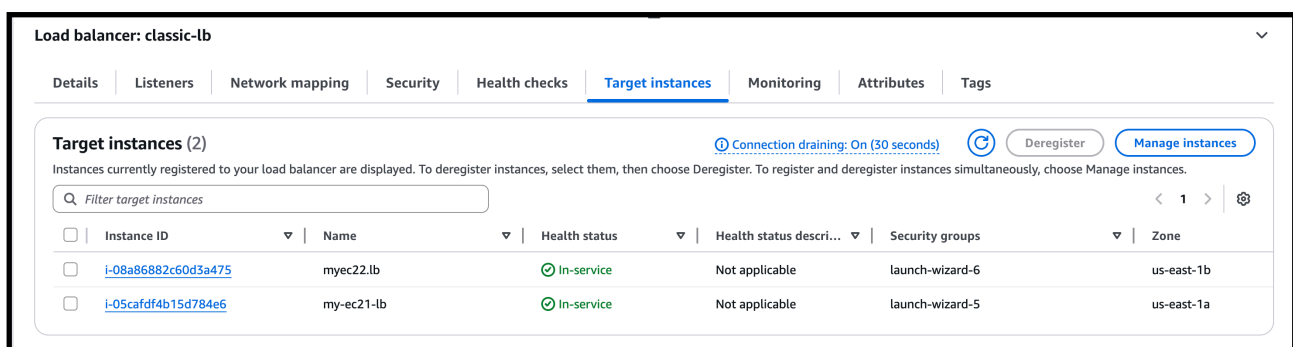
Step 7: Review & Create

- Review all settings
- Click **Create Load Balancer**



5 Validation Steps (With Evidence)

—> so target instances are in service which means we can now check the both instances are working in load balancer



—> copy the dns url of load balancer and paste in browser

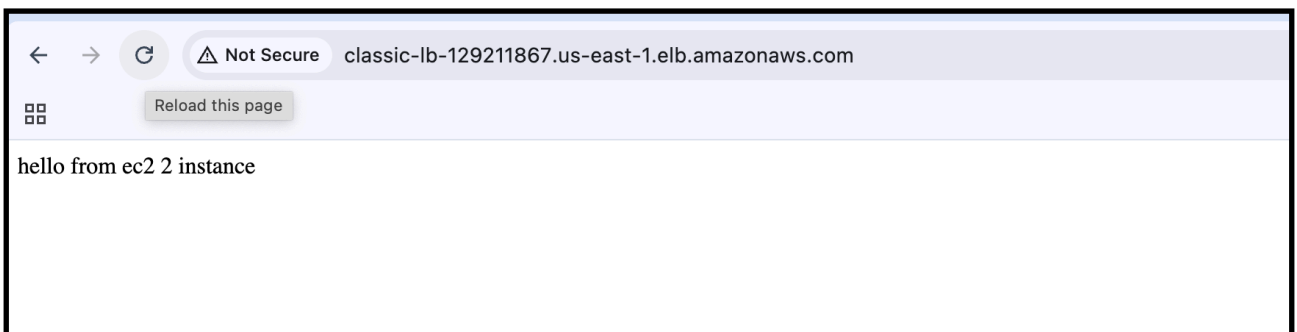
—> Functional Check

—> below it is showing the instance 1 index.html page



—>Traffic Distribution Check

Refresh the browser multiple times → traffic should switch between EC2s.



—> above is from ec2 2 instance

—> Conclusion

Classic Load Balancer was successfully created, configured, validated, and is now distributing traffic to EC2 instances as expected.

Task 2: Configure Application Load Balancer (ALB)

—> Task Title

Configure Application Load Balancer (ALB) in AWS

—> Objective

To create an Application Load Balancer that distributes HTTP traffic across EC2 instances using a target group and provides high availability across multiple Availability Zones.

—> Prerequisites

- ✓ VPC
- ✓ At least **2 Public Subnets** in different AZs
- ✓ EC2 instances running (web servers)
- ✓ Security Group for ALB (allow port 80)
- ✓ Security Group for EC2 (allow port 80 from ALB)
- ✓ Internet Gateway + Route Table
- ✓ Key Pair

—> Step-by-Step Implementation (With Evidence)

Step 1: Open ALB Creation Wizard

Console Path:

EC2 Dashboard → Load Balancing → **Load Balancers** → Create Load Balancer → **Application Load Balancer**

Explanation:

We select ALB because it works at Layer 7 and supports advanced routing.

EC2 > Load balancers > Compare and select load balancer type

Load balancer types

Application Load Balancer [Info](#)

Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

[Create](#)

Network Load Balancer [Info](#)

Choose a Network Load Balancer when you need ultra-high performance, TLS offloading at scale, centralized certificate deployment, support for UDP, and static IP addresses for your applications. Operating at the connection level, Network Load Balancers are capable of handling millions of requests per second securely while maintaining ultra-low latencies.

[Create](#)

Gateway Load Balancer [Info](#)

Choose a Gateway Load Balancer when you need to deploy and manage a fleet of third-party virtual appliances that support GENEVE. These appliances enable you to improve security, compliance, and policy controls.

[Create](#)

[Classic Load Balancer - previous generation](#)

Step 2: Configure Basic Load Balancer Settings

Fill the form:

Setting	Value
Name	Application-lb
Scheme	Internet Facing
IP Type	IPv4
Listener	HTTP (80)

Choose **2 public subnets** (must be in different AZs).

►

How Application Load Balancers work

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

application-lb

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme

Info

Scheme can't be changed after the load balancer is created.

☒ Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type

Info

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional

☒ IPv4

Includes only IPv4 addresses.

☐ Dualstack

Includes IPv4 and IPv6 addresses.

☐ Dualstack without public IPv4

Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with internet-facing load balancers only.

—-> before creating we have to create a target group in app-lb

Step 3: Create or Select Target Group

Target type: Instance

Name: application-tg

Protocol: HTTP

Port: 80

Health Check Path: /

Explanation:

Target group sends traffic to backend EC2 instances.

EC2

>

Target groups

>

Create target group

Create target group

Step 2 - recommended

Register targets

Step 3

Review and create

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

Target type

Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ Instances

Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.

Suitable for: ALB NLB GWLB

☐ IP addresses

Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.

Suitable for: ALB NLB GWLB

☐ Lambda function

Supports load balancing to a single Lambda function. ALB required as traffic source.

Suitable for: ALB

☐ Application Load Balancer

Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.

Suitable for: NLB

Target group name

Name must be unique per Region per AWS account.

application-tg

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 14/32

Protocol

Protocol for communication between the load balancer and targets.

HTTP

Port

Port number where targets receive traffic. Can be overridden for individual targets during registration.

80

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

☒ IPv4

Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

—-> attach vpc

☐

IPv6
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#) 

VPC
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-07968786753c2ea72 (default)
172.31.0.0/16

(default) ▼



Create VPC 

Protocol version

☒ HTTP1
Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

☐ HTTP2
Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

☐ gRPC
Send requests to targets using gRPC. Supported when the request protocol is gRPC.

- > attach both instances which few created in targets

80

1-65535 (separate multiple ports with commas)

Include as pending below

2 selections are now pending below. Include more or register targets when ready.

Review targets

Targets (2)

Remove all pending

☒ Show only pending

Instance ID

Name

Port

State

Security groups

Zone

Private IPv4 address

Subnet ID

Launch time

[i-08a86882c60d3a475](#)

myec22.lb

80

Running

launch-wizard-6

us-east-1b

172.31.84.198

subnet-0f208e7a1554613af

December 11, 2025,

[i-05cafd4b15d784e6](#)

my-ec21-lb

80

Running

launch-wizard-5

us-east-1a

172.31.6.144

subnet-071d209e29a1a1f82

December 11, 2025,

2 pending

Cancel

Previous

Next

—> after that review and create the target group

The screenshot shows the 'Create target group' wizard in the AWS Management Console. It is at Step 1: Target group details. The left sidebar shows three steps: Step 1: Create target group, Step 2: Register targets, and Step 3: Review and create. The main content area has a 'Review and create' header with a sub-header 'Review your target group configuration before creating'. Below this, there are two sections: 'Target group details' and 'Health check details'. The 'Target group details' section includes fields for Name (application-tg), Target type (Instance), Protocol : Port (HTTP: 80), Protocol version (HTTP1), VPC (vpc-07968786753c2ea72), and IP address type (IPv4). The 'Health check details' section includes fields for Health check protocol (HTTP), Health check path (/), Health check port (traffic-port), Interval (30 seconds), Timeout (5 seconds), Healthy threshold (5), Unhealthy threshold (2), and Success codes (200). There is an 'Edit' button at the top right of the details section. Below the details section is 'Step 2: Register targets' with an 'Edit' button. It shows a table of targets with 2 entries. The table has columns for Instance ID, Name, Port, and Zone. The first entry has Instance ID i-08a86882c60d3a475, Name myec22.lb, Port 80, and Zone us-east-1b. The second entry has Instance ID i-05c4fd4b15d784e6, Name my-ec21-lb, Port 80, and Zone us-east-1a.

Step 1: Create target group

Step 2 - recommended

Register targets

Step 3

Review and create

Review and create

Review your target group configuration before creating

Step 1: Target group details [Edit](#)

Target group details

Name	application-tg	Target type	Instance	Protocol : Port	HTTP: 80	Protocol version	HTTP1
VPC	vpc-07968786753c2ea72 Link						
IP address type	IPv4						

Health check details

Health check protocol	HTTP	Health check path	/	Health check port	traffic-port	Interval	30 seconds
Timeout	5 seconds	Healthy threshold	5	Unhealthy threshold	2	Success codes	200

Step 2: Register targets [Edit](#)

Targets (2)

Instance ID	Name	Port	Zone
i-08a86882c60d3a475	myec22.lb	80	us-east-1b
i-05c4fd4b15d784e6	my-ec21-lb	80	us-east-1a

—> now after creating target group lets directly select the group which we created

—> Select Target Group

Target type: Instance

Name: alb-target-group

Protocol: HTTP

Port: 80

Health Check Path: /

Explanation:

Target group sends traffic to backend EC2 instances.

The screenshot shows the 'Listeners and routing' page in the AWS Management Console. It has a header 'Listeners and routing' with an 'Info' link. Below the header is a description: 'A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.' There is a 'Remove' button at the top right. Below the description is a section for 'Listener HTTP:80'. It has a 'Protocol' dropdown set to 'HTTP' and a 'Port' input field set to '80'. Below this is a 'Default action' section with an 'Info' link and a description: 'The default action is used if no other rules apply. Choose the default action for traffic on this listener.' There are three radio buttons for 'Routing action': 'Forward to target groups' (selected), 'Redirect to URL', and 'Return fixed response'. Below the routing action is a 'Forward to target group' section with an 'Info' link and a description: 'Choose a target group and specify routing weight or [create target group](#).' There is a 'Target group' dropdown set to 'application-tg' and a 'Weight' input field set to '1'. Below the target group dropdown is a '+ Add target group' button and a description: 'You can add up to 4 more target groups.' There is a 'Target group stickiness' section with an 'Info' link and a description: 'Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.' There is a checkbox for 'Turn on target group stickiness' which is currently unchecked. At the bottom is a 'Listener tags - optional' section with a description: 'Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.'

Listeners and routing [Info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener HTTP:80 [Remove](#)

Protocol Port

Default action [Info](#)

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

☒ Forward to target groups ☐ Redirect to URL ☐ Return fixed response

Forward to target group [Info](#)

Choose a target group and specify routing weight or [create target group](#).

Target group Weight Percent

+ Add target group

You can add up to 4 more target groups.

Target group stickiness [Info](#)

Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

☐ Turn on target group stickiness

Listener tags - optional

Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

Step 3: Attach Security Group

- Select or create a security group
- Allow **Inbound: HTTP (80)**

Explanation:

ALB must accept traffic from the internet.

The screenshot shows the AWS IAM console page for a security group. The title is 'sg-0ffbfec998b25506f - default'. There are tabs for 'Inbound rules', 'Outbound rules', 'Sharing', 'VPC associations', and 'Tags'. The 'Inbound rules' tab is selected, showing a table with 4 rules. The details section at the top shows the security group name, ID, description, VPC ID, owner, and rule counts.

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-	sgr-02ff289ff1167363c	IPv4	All TCP	TCP	0 - 65535	0.0.0.0/0
-	sgr-00610f001e2a65ebe	IPv4	All traffic	All	All	0.0.0.0/0
-	sgr-0c9d17d94090b2bdc	-	All traffic	All	All	sg-0ffbfec998b25506f ...
-	sgr-00f2ae5af17030ec4	IPv4	SSH	TCP	22	0.0.0.0/0

Review and Create ALB

Click **Create Load Balancer**.

—> wait until status is active after it is active copy the the dns and check the browser

The screenshot shows the AWS EC2 console page for a newly created Application Load Balancer. The title is 'application-lb'. There are tabs for 'Listeners and rules', 'Network mapping', 'Resource map', 'Security', 'Monitoring', 'Integrations', 'Attributes', 'Capacity', and 'Tags'. The 'Listeners and rules' tab is selected, showing a table with 1 listener. The details section at the top shows the load balancer type, status, scheme, VPC, availability zones, load balancer IP address type, date created, load balancer ARN, and DNS name.

Protocol:Port	Default action	Rules	ARN	Security policy	Default SSL/TLS certificate	mTLS
-	-	-	-	-	-	-

Validation Steps (With Evidence)

✓ Validation 1: Target Group Health

Go to:

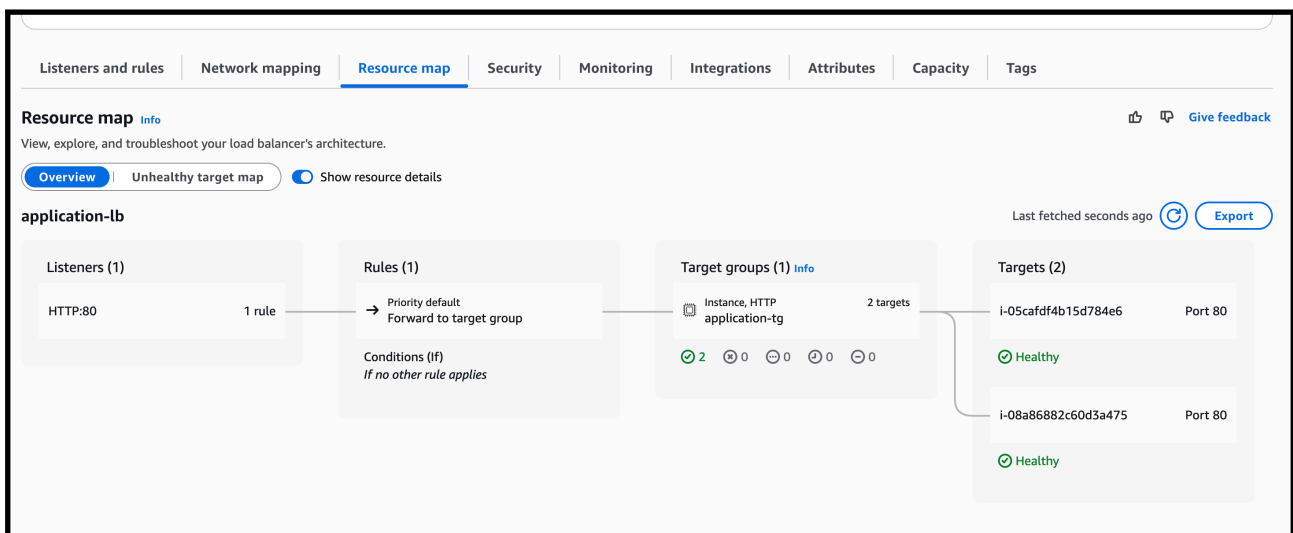
EC2 → Target Groups → Instances tab → Should show **healthy**.

—> also check the TG healthy checks of both instances

The screenshot shows the AWS Management Console for the 'application-tg' Target Group. The 'Instances' tab is active, displaying a summary of 2 healthy targets. Below this, the 'Registered targets' table lists two targets:

Instance ID	Name	Port	Zone	Health status	Health status details
i-08a86882c60d3a475	myec22.lb	80	us-east-1b (us...)	Healthy	-
i-05cfd4b15d784e6	my-ec21-lb	80	us-east-1a (us...)	Healthy	-

—> map of the load balance

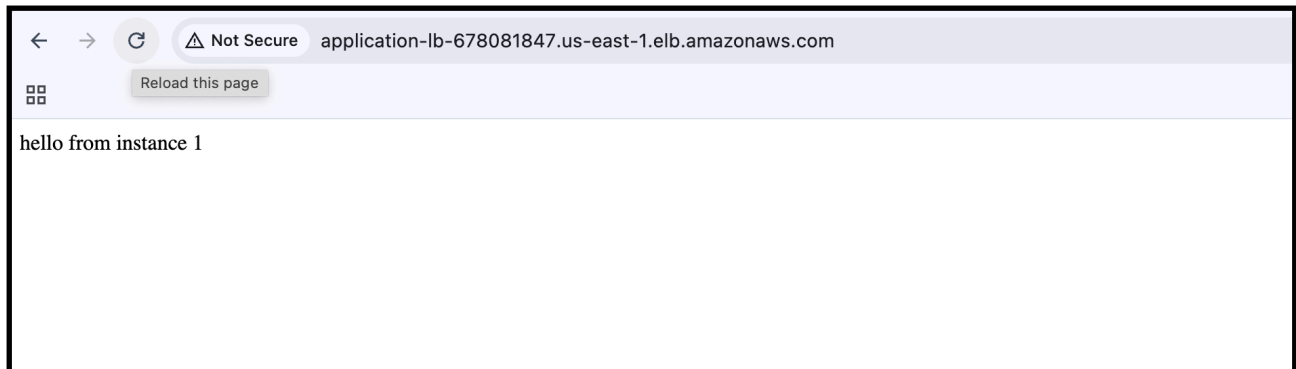


Validation 2: ALB DNS Functional Check

Open the ALB DNS in browser:

Expected: Backend webpage loads.

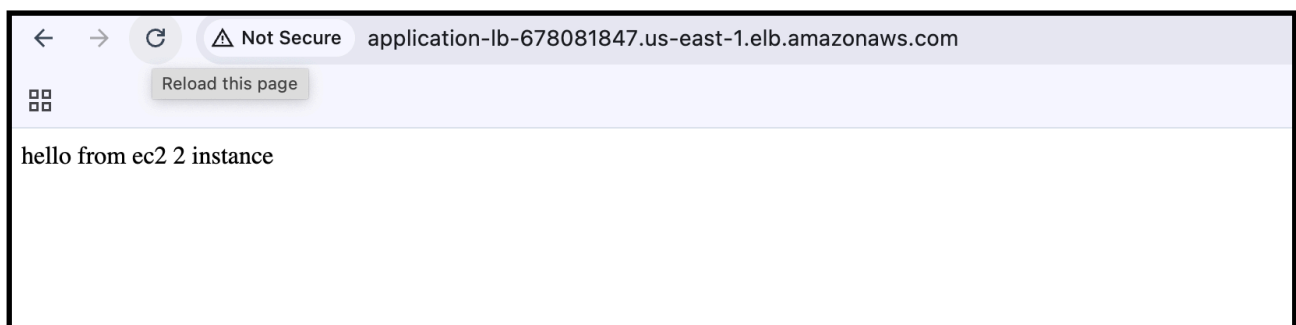
—> from ec2 1 instance



Validation 3: Load Distribution Check

Refresh the page repeatedly — traffic switches between EC2 servers.

—> from ec2 2 instance



Issues Faced (Optional)

- ALB not reachable → fixed by allowing port **80** in security group

—> given all traffic to allow all the traffics

sg-0ffbfc998b25506f - default

Actions

Details

Security group name

default

Security group ID

sg-0ffbfc998b25506f

Description

default VPC security group

VPC ID

vpc-07968786753c2ea72

Owner

207662791773

Inbound rules count

4 Permission entries

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Tags

Inbound rules (4)

Manage tags

Edit inbound rules

Search

Name

Security group rule ID

IP version

Type

Protocol

Port range

Source

-

sgr-02ff289ff1167363c

IPv4

All TCP

TCP

0 - 65535

0.0.0.0/0

-

sgr-00610f001e2a65ebe

IPv4

All traffic

All

All

0.0.0.0/0

-

sgr-0c9d17d94090b2bdc

-

All traffic

All

All

sg-0ffbfc998b25506f...

-

sgr-00f2ae5af17030ec4

IPv4

SSH

TCP

22

0.0.0.0/0

—>Conclusion

Application Load Balancer was successfully created, EC2 instances attached through a target group

EC2 > Load balancers

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMI

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Load balancers (1/2)

What's new?

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

Name

State

Type

Scheme

IP address type

VPC ID

Availability Zones

Security groups

classic-lb

-

classic

-

vpc-07968786753c2ea72

2 Availability Zones

sg-0ffbfc998b25

application-lb

Active

application

Internet-facing

IPv4

vpc-07968786753c2ea72

2 Availability Zones

sg-0ffbfc998b25

Task 3: Configure Network Load Balancer (NLB)

—>Task Title

Configure Network Load Balancer (NLB) in AWS

—> Objective

To create a **Network Load Balancer** that routes TCP traffic to backend EC2 instances with ultra-low latency and high performance at Layer 4.

—> Prerequisites

- ✓ VPC
- ✓ At least **2 public subnets** in different AZs
- ✓ EC2 instances running (web servers / TCP servers)
- ✓ Security Group for EC2 allowing **TCP 80**
(*NLB does NOT use a security group*)
- ✓ Internet Gateway + Route Tables
- ✓ IAM permissions to create NLB
- ✓ Key Pair for EC2 login

—>Step-by-Step Implementation (With Evidence)

Step 1: Open the NLB Creation Wizard

Console Path:

EC2 Dashboard → Load Balancers → **Create Load Balancer** → Select **Network Load Balancer**

Explanation:

NLB operates at **Layer 4 (TCP/UDP)** and provides static IP with very high performance.

EC2 > Load balancers > Compare and select load balancer type

Load balancer types

Application Load Balancer [Info](#)

Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

[Create](#)

Network Load Balancer [Info](#)

Choose a Network Load Balancer when you need ultra-high performance, TLS offloading at scale, centralized certificate deployment, support for UDP, and static IP addresses for your applications. Operating at the connection level, Network Load Balancers are capable of handling millions of requests per second securely while maintaining ultra-low latencies.

[Create](#)

Gateway Load Balancer [Info](#)

Choose a Gateway Load Balancer when you need to deploy and manage a fleet of third-party virtual appliances that support GENEVE. These appliances enable you to improve security, compliance, and policy controls.

[Create](#)

[Classic Load Balancer - previous generation](#)

Step 2: Configure Basic Load Balancer Settings

Fill the following details:

Setting	Value
Name	Network-lb
Scheme	Internet-facing
IP Type	IPv4
Listener	TCP (80)

Select **two public subnets** (different AZs).

Create Network Load Balancer [Info](#)

The Network Load Balancer distributes incoming TCP and UDP traffic across multiple targets such as Amazon EC2 instances, microservices, and containers. When the load balancer receives traffic on the port that are specified in the listener configuration, and the routing rule specified as the default action.

► How Network Load Balancers work

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

network-lb

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme

Scheme can't be changed after the load balancer is created.

☒ Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.

Load balancer IP address type [Info](#)

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types.

☒ IPv4

Includes only IPv4 addresses.

☐ Dualstack

Includes IPv4 and IPv6 addresses.

Network mapping [Info](#)

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to on-premises targets or if using VPC peering.

—> select the vpc and two subnets zones

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to on-premises targets or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-07968786753c2ea72 (default)
172.31.0.0/16

Availability Zones and subnets

Select one or more Availability Zones and corresponding subnets. Enabling multiple Availability Zones increases the fault tolerance of your applications. The load balancer or the VPC are not available for selection.

☒ us-east-1a (use1-az1)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-071d209e29a1a1f82
IPv4 subnet CIDR: 172.31.0.0/20

IPv4 address

The front-end IPv4 address of the load balancer in the selected Availability Zone.

☒ Assigned by AWS

☐ Use an Elastic IP

☒ us-east-1b (use1-az2)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0f208e7a1554613af
IPv4 subnet CIDR: 172.31.80.0/20

IPv4 address

The front-end IPv4 address of the load balancer in the selected Availability Zone.

☒ Assigned by AWS

☐ Use an Elastic IP

☐ us-east-1c (use1-az4)

☐ us-east-1d (use1-az6)

☐ us-east-1e (use1-az3)

☐ us-east-1f (use1-az5)

—> similarly go to target group and after creating the tg come to same page and continue creating the Network lb

Step 3: Create Target Group for NLB

Target group settings:

Field	Value
Target Type	Instance
Name	nlb-target-group
Protocol	TCP
Port	80

ec2—>target group—>create target group

EC2 > Target groups > Create target group

Step 1

Create target group

Step 2 - recommended

Register targets

Step 3

Review and create

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation."

Target type

Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

Instances

Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.

Suitable for: ALB NLB GWLB

IP addresses

Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.

Suitable for: ALB NLB GWLB

Lambda function

Supports load balancing to a single Lambda function. ALB required as traffic source.

Suitable for: ALB

Application Load Balancer

Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.

Suitable for: NLB

Target group name

Name must be unique per Region per AWS account.

network-tg

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 10/32

Protocol

Protocol for communication between the load balancer and targets.

TCP

Port

Port number where targets receive traffic. Can be overridden for individual targets during registration.

80

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

—> select the vpc and health check protocol to tcp

the one that will be applied to the target.

IPv6

Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-07968786753c2ea72 (default)

172.31.0.0/16

(default)

Create VPC

Health checks

The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

Health check protocol

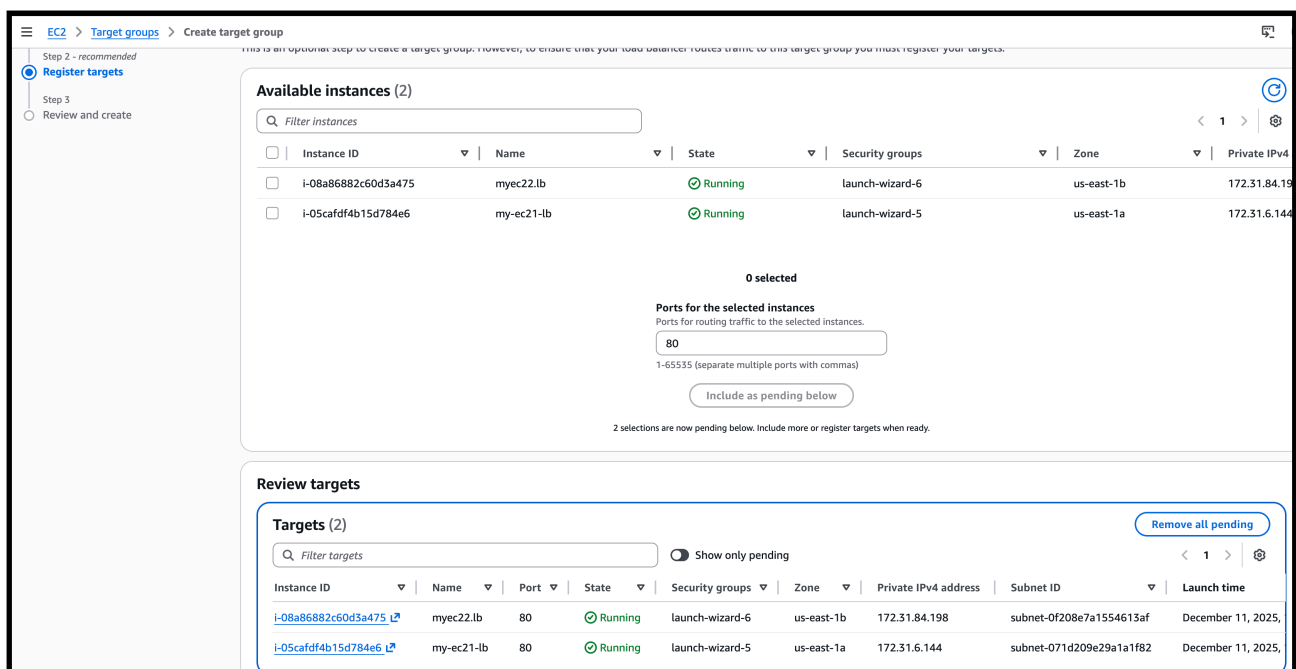
TCP

► Advanced health check settings

Register EC2 Instances to Target Group

- Select EC2 instances
- Click **Include as pending below**
- Click **Create Target Group**

—> and add the ec2 instance which we create and target it



Step 2 - recommended

Register targets

Step 3

Review and create

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (2)

Filter instances

☐	Instance ID	Name	State	Security groups	Zone	Private IPv4
☐	i-08a86882c60d3a475	myec22-lb	Running	launch-wizard-6	us-east-1b	172.31.84.198
☐	i-05c4df4b15d784e6	my-ec21-lb	Running	launch-wizard-5	us-east-1a	172.31.6.144

0 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

Include as pending below

2 selections are now pending below. Include more or register targets when ready.

Review targets

Targets (2)

Filter targets

Show only pending

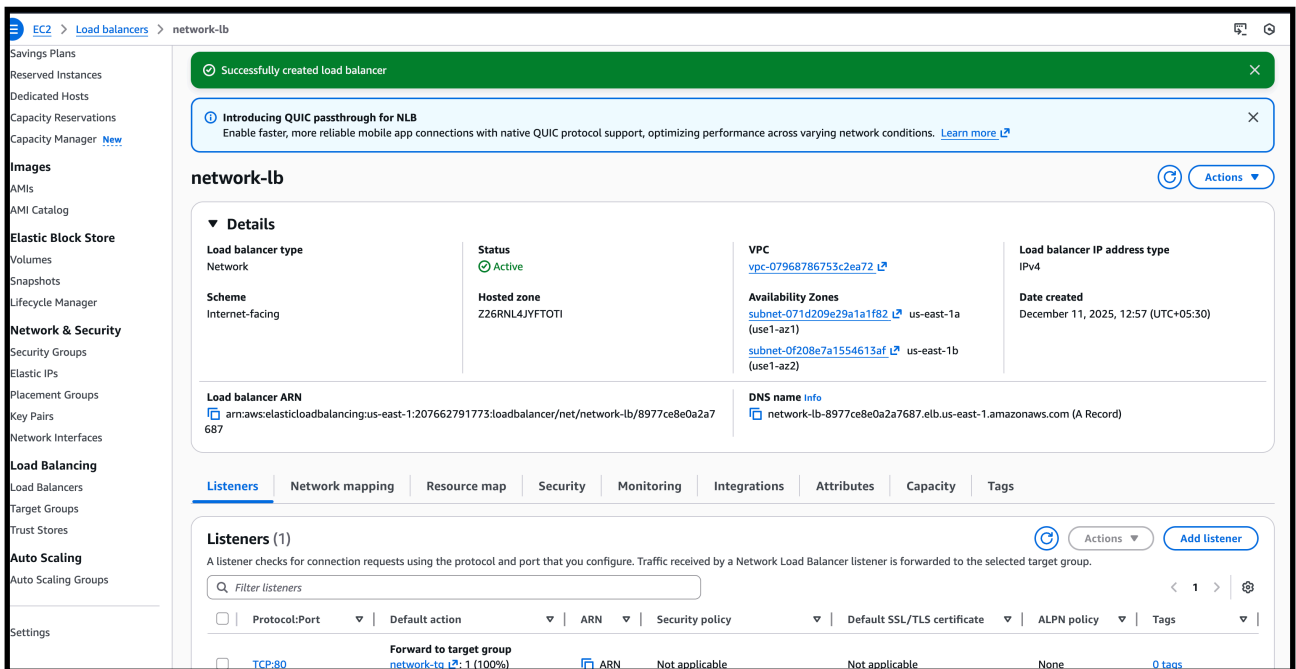
Remove all pending

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch time
i-08a86882c60d3a475	myec22-lb	80	Running	launch-wizard-6	us-east-1b	172.31.84.198	subnet-0f208e7a1554613af	December 11, 2025,
i-05c4df4b15d784e6	my-ec21-lb	80	Running	launch-wizard-5	us-east-1a	172.31.6.144	subnet-071d209e29a1a1f82	December 11, 2025,

—> after creating the target group attach the target group click next

Step 4: Review & Create Network Load Balancer

- Review all settings
- Click **Create Load Balancer**
- Wait until NLB status = **Active**

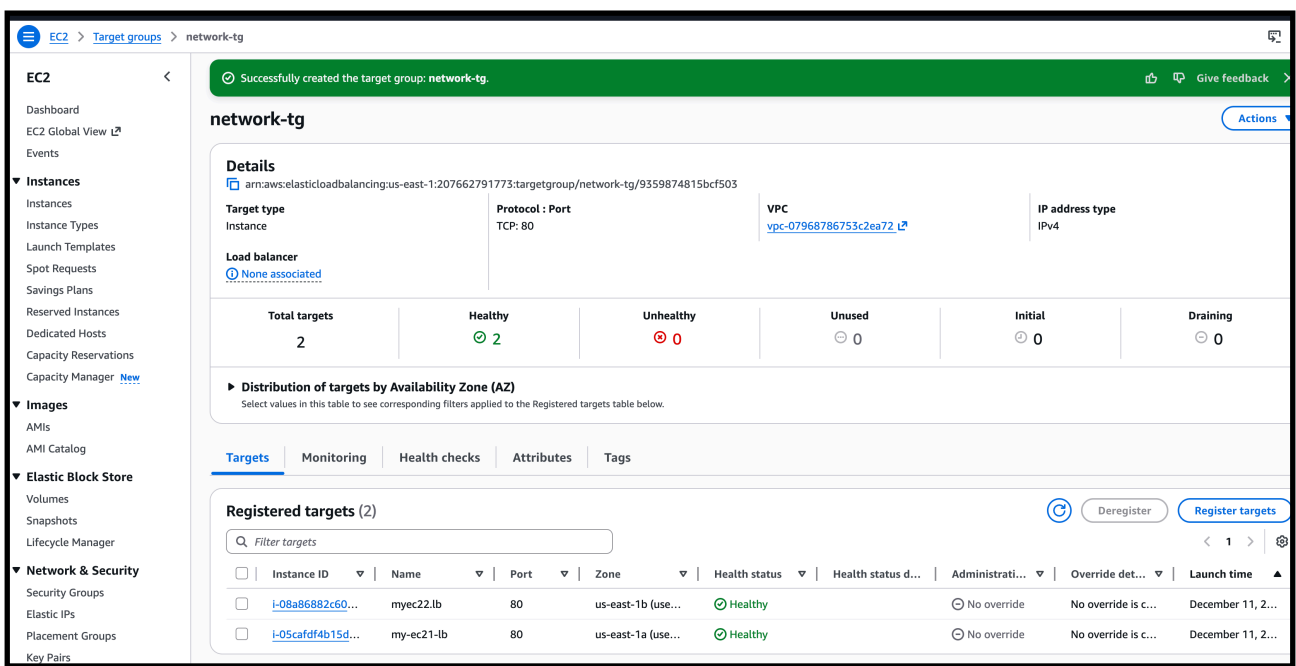


Validation Steps (With Evidence)

✓ Validation 1: Target Group Health

Path: EC2 → Target Groups → Instances tab
Expected: **Healthy**

— → check both target groups whether it is healthy and also check the NLB is active



Validation 2: NLB DNS Functional Check

Copy NLB DNS:



—->issues Faced :~

Issue	Fix
Target group unhealthy	Ensured EC2 SG allowed TCP 80

— —> added the tcp protocol to security group

sg-0ffbfec998b25506f - default

Actions

Details

Security group name
default

Owner
207662791773

Security group ID
sg-0ffbfec998b25506f

Inbound rules count
4 Permission entries

Description
default VPC security group

Outbound rules count
1 Permission entry

VPC ID
vpc-07968786753c2ea72

Inbound rules

Outbound rules

Sharing

VPC associations

Tags

Inbound rules (4)

Search

Manage tags

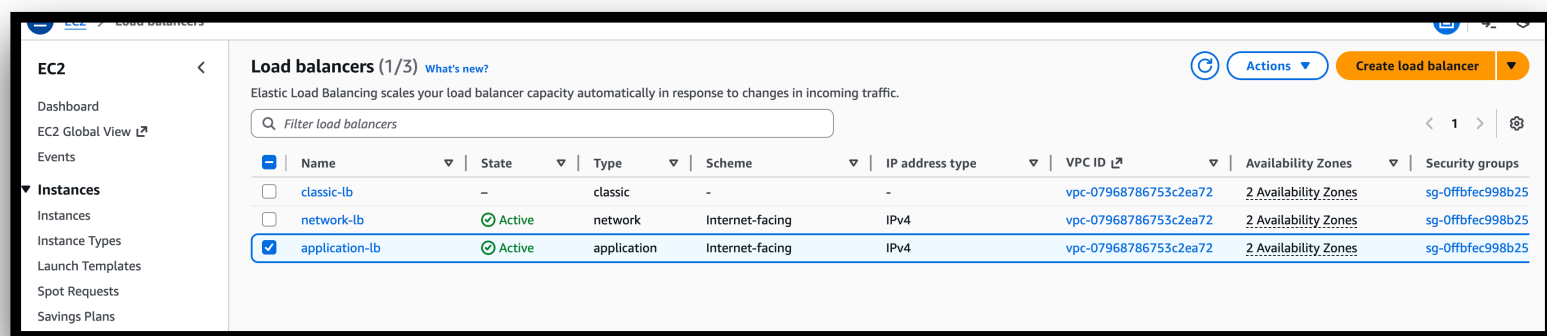
Edit inbound rules

	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
☐	-	sgr-02ff289ff1167363c	IPv4	All TCP	TCP	0 - 65535	0.0.0.0/0
☐	-	sgr-00610f001e2a65ebe	IPv4	All traffic	All	All	0.0.0.0/0
☐	-	sgr-0c9d17d94090b2bdc	-	All traffic	All	All	sg-0ffbfec998b25506f...
☐	-	sgr-00f2ae5af17030ec4	IPv4	SSH	TCP	22	0.0.0.0/0

Conclusion

Network Load Balancer was successfully created, EC2 instances were attached using a TCP target group, and traffic is now load balanced with high performance at Layer 4.

—>Task 1, task 2, task3 all types of load balancers created



—>Task 4: Attach SSL Certificate to Application Load Balancer (ALB)

1 Task Title

Attach SSL Certificate (HTTPS) to Application Load Balancer using AWS ACM

2 Objective

To enable **HTTPS (port 443)** on the Application Load Balancer by attaching an SSL certificate issued from AWS Certificate Manager (ACM), ensuring secure encrypted communication between clients and the ALB.

3 Prerequisites

- ✓ Application Load Balancer (ALB) already created

- ✓ Target group configured & healthy
- ✓ Domain name purchased (GoDaddy / Namecheap / Route 53)
- ✓ Hosted Zone in Route 53 (optional if using external DNS)
- ✓ SSL certificate issued and **validated** in ACM
 - Certificate **must be in the same region as ALB**
 - If using CloudFront, certificate must be in us-east-1
- ✓ ALB security group must allow **HTTPS (443)**
- ✓ EC2 security group must allow inbound from ALB on required port (80)

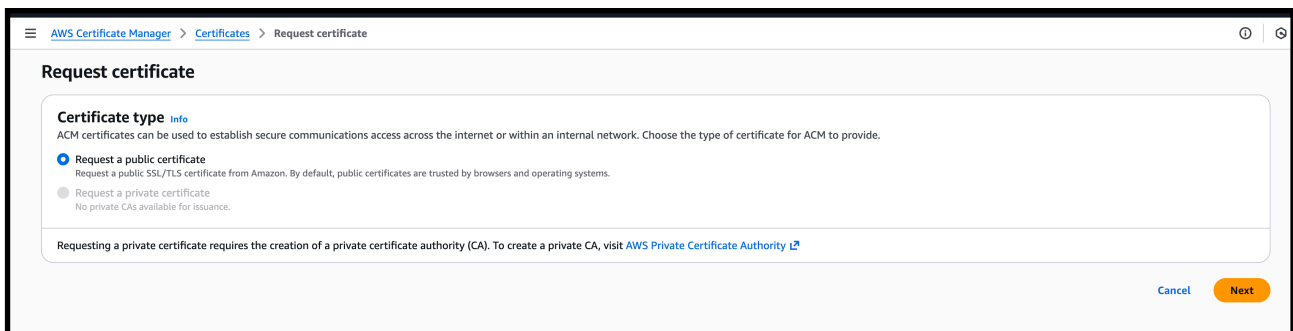
—>Step-by-Step Implementation (With Evidence)

Step 1: Generate SSL Certificate in ACM

Console Path:

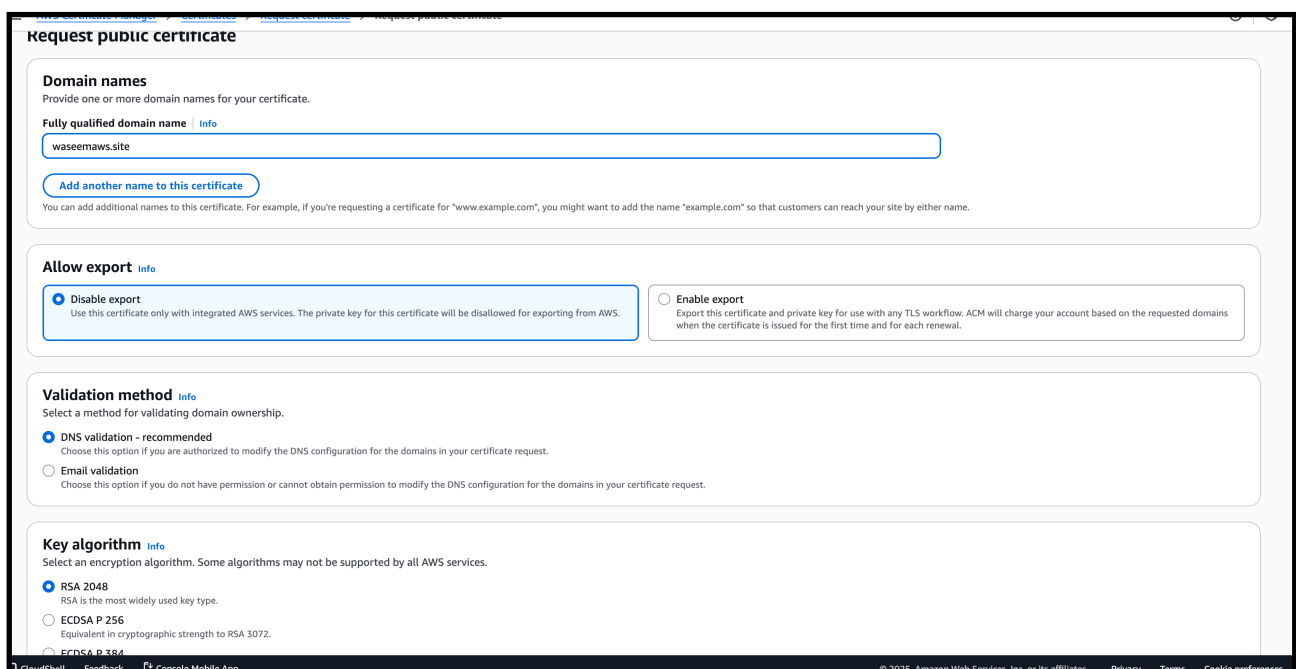
AWS Management Console → Certificate Manager → **Request a Certificate**

Choose: **Request a Public Certificate**



The screenshot shows the 'Request certificate' page in the AWS Certificate Manager console. The breadcrumb trail is 'AWS Certificate Manager > Certificates > Request certificate'. The page title is 'Request certificate'. Under 'Certificate type', there are two options: 'Request a public certificate' (selected with a blue radio button) and 'Request a private certificate' (unselected with a grey radio button). Below the public certificate option, it says 'Request a public SSL/TLS certificate from Amazon. By default, public certificates are trusted by browsers and operating systems.' Below the private certificate option, it says 'No private CAs available for issuance.' A note at the bottom states: 'Requesting a private certificate requires the creation of a private certificate authority (CA). To create a private CA, visit [AWS Private Certificate Authority](#).' At the bottom right, there are 'Cancel' and 'Next' buttons.

—> enter the domain name which you purchased



The screenshot shows the 'request public certificate' page in the AWS Certificate Manager console. The breadcrumb trail is 'AWS Certificate Manager > Certificates > Request certificate > request public certificate'. The page title is 'request public certificate'. Under 'Domain names', it says 'Provide one or more domain names for your certificate.' There is a text input field for 'Fully qualified domain name' with the value 'waseemaws.site'. Below the input field is a button 'Add another name to this certificate'. A note below the button says: 'You can add additional names to this certificate. For example, if you're requesting a certificate for "www.example.com", you might want to add the name "example.com" so that customers can reach your site by either name.' Under 'Allow export', there are two options: 'Disable export' (selected with a blue radio button) and 'Enable export' (unselected with a grey radio button). Below the 'Disable export' option, it says 'Use this certificate only with integrated AWS services. The private key for this certificate will be disallowed for exporting from AWS.' Below the 'Enable export' option, it says 'Export this certificate and private key for use with any TLS workflow. ACM will charge your account based on the requested domains when the certificate is issued for the first time and for each renewal.' Under 'Validation method', it says 'Select a method for validating domain ownership.' There are two options: 'DNS validation - recommended' (selected with a blue radio button) and 'Email validation' (unselected with a grey radio button). Below the 'DNS validation' option, it says 'Choose this option if you are authorized to modify the DNS configuration for the domains in your certificate request.' Below the 'Email validation' option, it says 'Choose this option if you do not have permission or cannot obtain permission to modify the DNS configuration for the domains in your certificate request.' Under 'Key algorithm', it says 'Select an encryption algorithm. Some algorithms may not be supported by all AWS services.' There are three options: 'RSA 2048' (selected with a blue radio button), 'ECDSA P 256' (unselected with a grey radio button), and 'ECDSA P 384' (unselected with a grey radio button). Below the 'RSA 2048' option, it says 'RSA is the most widely used key type.' Below the 'ECDSA P 256' option, it says 'Equivalent in cryptographic strength to RSA 3072.' At the bottom left, there are links for 'CloudShell', 'Feedback', and 'Console Mobile App'. At the bottom right, there is a copyright notice '© 2025, Amazon Web Services, Inc. or its affiliates.' and links for 'Privacy', 'Terms', and 'Cookie preferences'.

—> my domain name is waseemaws.site

1. Select **DNS Validation**
2. Click **Request**

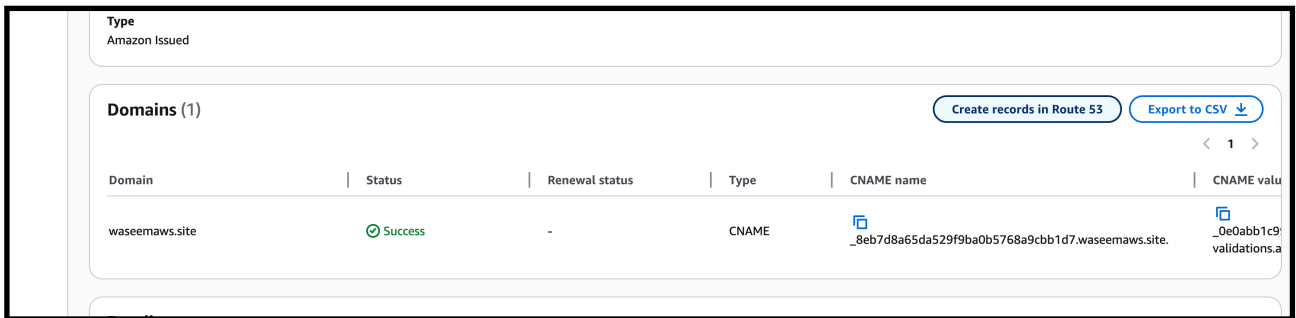
Step 2: Create DNS Records in Route 53

After requesting:

1. ACM will show a **CNAME record** required for validation.
2. Click the button:
✓ **Create records in Route 53**
3. AWS automatically creates a CNAME validation record inside the correct hosted zone.

—> below we can see there is create record in route 53

(∴ the image is before success so the moment we be creating the route53 record the status will be in progress



The screenshot shows the AWS Route 53 console interface. At the top, it indicates the 'Type' is 'Amazon Issued'. Below this, there's a section for 'Domains (1)' with two buttons: 'Create records in Route 53' and 'Export to CSV'. A table below lists the domain 'waseemaws.site' with a status of 'Success', a renewal status of '-', and a CNAME type. The CNAME name is '_8eb7d8a65da529f9ba0b5768a9cbb1d7.waseemaws.site.' and the CNAME value is '_0e0abb1c9validations.a'.

Domain	Status	Renewal status	Type	CNAME name	CNAME value
waseemaws.site	Success	-	CNAME	_8eb7d8a65da529f9ba0b5768a9cbb1d7.waseemaws.site.	_0e0abb1c9validations.a

Step 3: Create the Hosted Zone in Route 53 (You Forgot This Step Earlier)

1. I went to **Route 53** → **Hosted Zones**
2. Clicked **Create Hosted Zone**

Route 53 > Hosted zones > Create hosted zone

Create hosted zone [Info](#)

Hosted zone configuration
A hosted zone is a container that holds information about how you want to route traffic for a domain, such as example.com, and its subdomains.

Domain name [Info](#)
This is the name of the domain that you want to route traffic for.

Valid characters: a-z, 0-9, ! " # \$ % & ' () * + , - / : ; < = > ? @ [\] ^ _ ` { | } . ~

Description - optional [Info](#)
This value lets you distinguish hosted zones that have the same name.

The description can have up to 256 characters. 0/256

Type [Info](#)
The type indicates whether you want to route traffic on the internet or in an Amazon VPC.

☒ **Public hosted zone**
A public hosted zone determines how traffic is routed on the internet.

☐ **Private hosted zone**
A private hosted zone determines how traffic is routed within an Amazon VPC.

Tags [Info](#)
Apply tags to hosted zones to help organize and identify them.

No tags associated with the resource.

[Add tag](#)

- Entered:
`waseemaws.site`
- Route 53 generated **NS** and **SOA** records automatically.
- And later copied these **4 NS records** from Route 53.

Route 53 > Hosted zones > waseemaws.site

Route 53

- Dashboard
- Hosted zones
- Health checks
- Profiles
- Global Resolver
 - Global resolvers [New](#)
- VPC Resolver
 - VPCs
 - Inbound endpoints
 - Outbound endpoints
 - Rules
 - Query logging
 - Outposts
- Domains
 - Registered domains
 - Requests
- IP-based routing
 - CIDR collections
- Traffic flow
 - Traffic policies
 - Policy records

Hosted zones

Public **waseemaws.site** [Info](#) [Delete zone](#) [Test record](#) [Configure query logging](#)

Hosted zone details [Edit hosted zone](#)

Records (2) [Info](#) [Delete record](#) [Import zone file](#) [Create record](#)

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

[Type](#) [Routing p...](#) [Alias](#) [1](#) [Settings](#)

☐	Record name	Type	Routin...	Differ...	Alias	Value/Route traffic to	TTL (s...)
☐	waseemaws.site	NS	Simple	-	No	ns-156.awsdns-19.com. ns-1129.awsdns-13.org. ns-906.awsdns-49.net. ns-1636.awsdns-12.co.uk.	172800
☐	waseemaws.site	SOA	Simple	-	No	ns-156.awsdns-19.com. awsd...	900

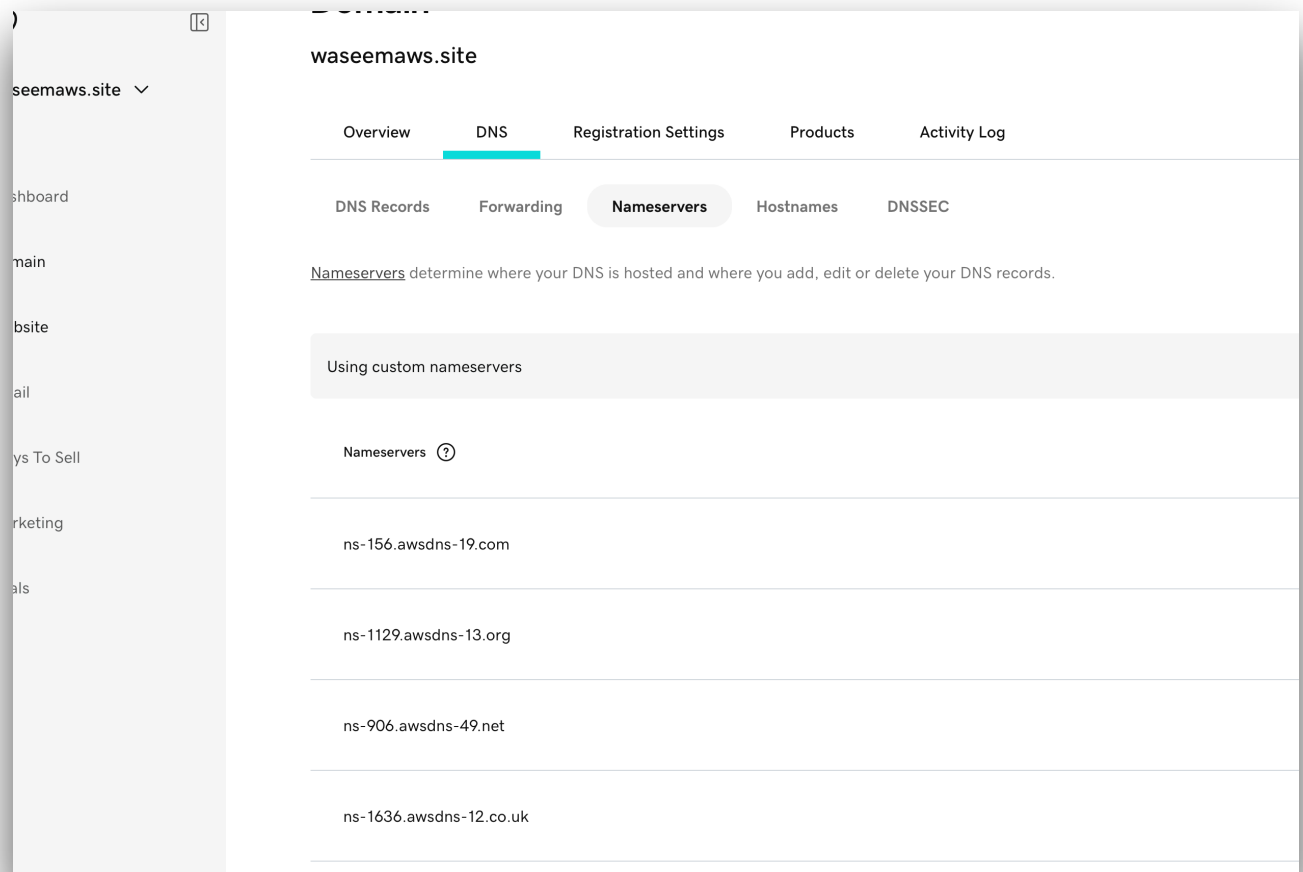
0 records
Select a...

—-> when we used step 2 at that moment we created a cname automatically

☐	waseemaws.site	NS	Simple	-	No	ns-156.awsdns-19.com. ns-1129.awsdns-13.org. ns-906.awsdns-49.net. ns-1636.awsdns-12.co.uk.	172800
☐	waseemaws.site	SOA	Simple	-	No	ns-156.awsdns-19.com. awsd...	900
☐	_8eb7d8a65da52...	CNAME	Simple	-	No	_0e0abb1c99c37c52a2d6df0...	300

Step 4: Update Nameservers in GoDaddy

—-> added those names in go daddy site



—> Add this name servers in side the dns

— — And simply create the hosted zone of it and after creating in hosted zones we will get the

—-> server names of the DNS and we have to copy this names and paste in GoDaddy site were we purchased the domain

ns-156.awsdns-19.com.
ns-1129.awsdns-13.org.
ns-906.awsdns-49.net.
ns-1636.awsdns-12.co.uk

Step 5: ACM Automatically Validated the Certificate

After global DNS propagation:

1. The CNAME record from ACM became active
2. ACM detected the validation
3. Status changed to:

Success
Issued

The screenshot displays the AWS Certificate Manager (ACM) console. The main content area shows the details for a certificate with the identifier `cd5f7360-8b0f-4f73-8dbf-5391780e0391`. The certificate status is **Issued**, indicated by a green checkmark icon. The domains listed for this certificate include `waseemaws.site`, which has a **Success** validation status. The details section shows the certificate is **In use**, has a **Serial number** of `02:4a:cf:fa:fa:6f:2c:e0:e0:ba:0e:d3:8b:02:f4:40`, was **Requested at** December 11, 2025, 16:14:02 (UTC+05:30), and is **Renewal eligible** (Ineligible).

Domain	Status	Renewal status	Type	CNAME name	CNAME value
waseemaws.site	Success	-	CNAME	<code>_8eb7d8a65da529f9ba0b5768a9cbb1d7.waseemaws.site.</code>	<code>_0e0abb1c9validations.a</code>

You now have a valid SSL certificate for `waseemaws.site`.

Step 6: Open the ALB Listener Settings

Console Path:

EC2 → Load Balancers → Select your ALB → **Listeners** tab → Add Listener

Choose:

- Protocol: **HTTPS**
- Port: **443**
- Default action: Forward to your target group

The screenshot shows the AWS Management Console interface for an Application Load Balancer (ALB). The left sidebar contains navigation links for various AWS services. The main content area displays the 'Details' tab for the ALB, showing its status as 'Active', VPC ID, Availability Zones, and Load balancer IP address type. Below this, the 'Listeners and rules' tab is selected, showing a single listener for HTTP:80. The listener is configured to forward traffic to the target group 'application-tg' with a weight of 1 (100%). The target group stickiness is set to 'Off'.

—> Fill the protocol with https

—> port : 443

The screenshot shows the 'Listener: HTTPS:443' configuration page in the AWS Management Console. The 'Protocol' is set to HTTPS and the 'Port' is 443. The 'Default action' is 'Forward to target group'. The 'Pre-routing action' is 'No pre-routing action'. The 'Routing action' is 'Forward to target group'. The 'Target group' is 'application-tg' with a weight of 1 (100%). The 'Target group stickiness' is set to 'Off'.

—> Attach SSL Certificate to ALB

Under **Secure Listener (HTTPS 443)** → Choose ACM certificate:

- Select your domain certificate from ACM
- Choose **Security Policy**

Secure listener settings [Info](#)

Security policy [Info](#)

Your load balancer uses a Secure Socket Layer (SSL) negotiation configuration called a security policy to manage SSL connections with clients. [Compare security policies](#)

Security category

All security policies

Policy name [New](#)

ELBSecurityPolicy-TLS13-1-2-Res-PQ-2025-09 (Recommended)

Default SSL/TLS server certificate

The certificate used if a client connects without SNI protocol, or if there are no matching certificates. You can source this certificate from AWS Certificate Manager (ACM), Amazon Identity and Access Management (IAM), or import a certificate. This certificate will automatically be added to your listener certificate list.

Certificate source

☒ From ACM

☐ From IAM

☐ Import certificate

Certificate (from ACM)

The selected certificate will be applied as the default SSL/TLS server certificate for this load balancer's secure listeners.

waseemaws.site

cd5f7360-8b0f-4f73-8dbf-5391780e0391

[Request new ACM certificate](#)

Client certificate handling [Info](#)

Client certificates are used to make authenticated requests to remote servers. [Learn more](#)

☐ Mutual authentication (mTLS)

Mutual TLS (Transport Layer Security) authentication offers two-way peer authentication. It adds a layer of security over TLS and allows your services to verify the client that's making the connection.

Default attributes are applied when adding a listener. You can edit them after adding the listener.

Step 7: Save Listener Configuration

Click **Save** to apply HTTPS settings.

ALB now supports:

- HTTP (80)
- HTTPS (443)

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Listeners and rules

Network mapping

Resource map

Security

Monitoring

Integrations

Attributes

Capacity

Tags

Listeners and rules (2) Info

A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.

Filter listeners

< 1 >

☐	Protocol:Port	Default action	Rules	ARN	Security policy	Default SSL/TLS certificate	mTLS
☐	HTTP:80	• Forward to target group application-tg 1 (100%) Target group stickiness: Off	1 rule	ARN	Not applicable	Not applicable	Not applicable
☐	HTTPS:443	• Forward to target group application-tg 1 (100%) Target group stickiness: Off	1 rule	ARN	ELBSecurityPolicy-TLS13-1-2-...	waseemaws.site (Certificate ID: ...)	Off

Step 8:

—-> Update ALB Security Group

We have already added the tcp and https in our sg

Type	Protocol	Port	Source
HTTPS	TCP	443	0.0.0.0/

Inbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Source	Security groups
	sgr-020f51bb3242ab4ea	22	TCP	0.0.0.0/0	launch-wizard-6
	sgr-07c44de8817c8c76f	All	All	0.0.0.0/0	launch-wizard-6
	sgr-073ef0fd6b6605ed7	0 - 65535	TCP	0.0.0.0/0	launch-wizard-6

Outbound rules

—> task 5 Map Application load balancer to R53

Route 53 → Hosted Zone → Create Record

Then you created:

Record type: A

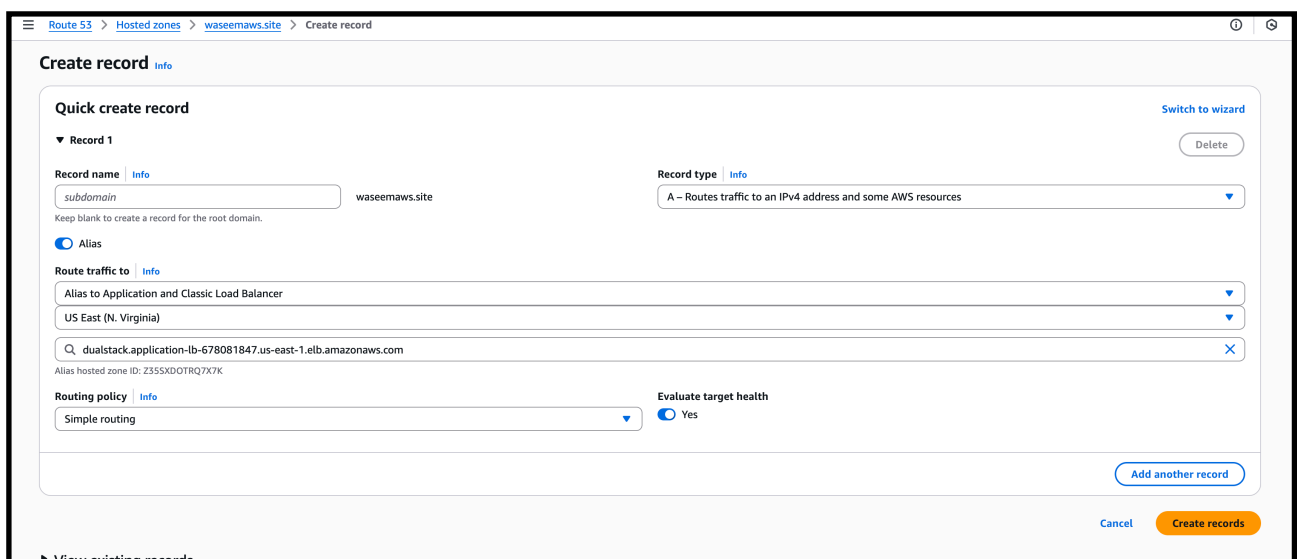
Record name: waseemaws.site

Alias: Yes

Alias target: <Your Application Load Balancer DNS Name>

Routing policy: Simple

This connects: Your domain → ALB → EC2 instances



The screenshot shows the 'Create record' page in the AWS Route 53 console. The breadcrumb trail at the top is 'Route 53 > Hosted zones > waseemaws.site > Create record'. The page title is 'Create record' with an 'info' link. Below the title is a 'Quick create record' section with a 'Switch to wizard' link. Under 'Record 1', there is a 'Record name' field with a placeholder 'subdomain' and a value 'waseemaws.site'. A note says 'Keep blank to create a record for the root domain.' To the right is a 'Record type' dropdown menu set to 'A - Routes traffic to an IPv4 address and some AWS resources'. Below this is a radio button for 'Alias' which is selected. The 'Route traffic to' section has a dropdown for 'Alias to Application and Classic Load Balancer', a dropdown for 'US East (N. Virginia)', and a text input field containing 'dualstack.application-lb-678081847.us-east-1.elb.amazonaws.com'. Below the text input is the 'Alias hosted zone ID: Z35SXDOTRQ7X7K'. The 'Routing policy' dropdown is set to 'Simple routing'. There is a toggle for 'Evaluate target health' which is turned 'Yes'. At the bottom right are buttons for 'Add another record', 'Cancel', and 'Create records'. A link 'View existing records' is at the bottom left.

Your domain now works with HTTPS

Because:

- SSL is active

Records (5)

Accelerated recovery

DNSSEC signing

Hosted zone tags (0)

Records (5) Info

Delete record

Import zone file

Create record

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Q

Filter records by property or value

Type

Routing p...

Alias

<

1

>

☐	Record name	Type	Routin...	Differ...	Alias	Value/Route traffic to	TTL (s...
☐	waseemaws.site	A	Simple	-	Yes	dualstack.application-lb-678...	-
☐	waseemaws.site	NS	Simple	-	No	ns-156.awsdns-19.com. ns-1129.awsdns-13.org. ns-906.awsdns-49.net. ns-1636.awsdns-12.co.uk.	172800
☐	waseemaws.site	SOA	Simple	-	No	ns-156.awsdns-19.com. awsd...	900
☐	_8eb7d8a65da52...	CNAME	Simple	-	No	_0e0abb1c99c37c52a2d6df0...	300
☐	_8eb7d8a65da52...	CNAME	Simple	-	No	_0e0abb1c99c37c52a2d6df0...	30

Domain routes to ALB

- ALB forwards traffic to EC2

—> Validation 1:

Open the HTTPS URL: : waseemaws.site

Untitled 1 < >

waseemaws.site

★ Waseem's DevOps Toolbox

Your single-page DevOps navigation dashboard (CI/CD | Infra | Monitoring | Containers | Security)

Search tools like Jenkins, Docker, Grafana...

All Categories

Jenkins

Open-source CI/CD automation

Open

GitHub Actions

GitHub-native CI/CD

Open

GitLab

DevOps lifecycle platform

Open

Terraform

Infrastructure as Code

Open

Ansible

Config automation

Open

Docker

Container E

Open

Kubernetes

Container orchestration

Open

Prometheus

Monitoring system

Open

Grafana

Dashboards & visualization

Open

ELK Stack

Logging & analytics

Open

Vault

Secrets manager

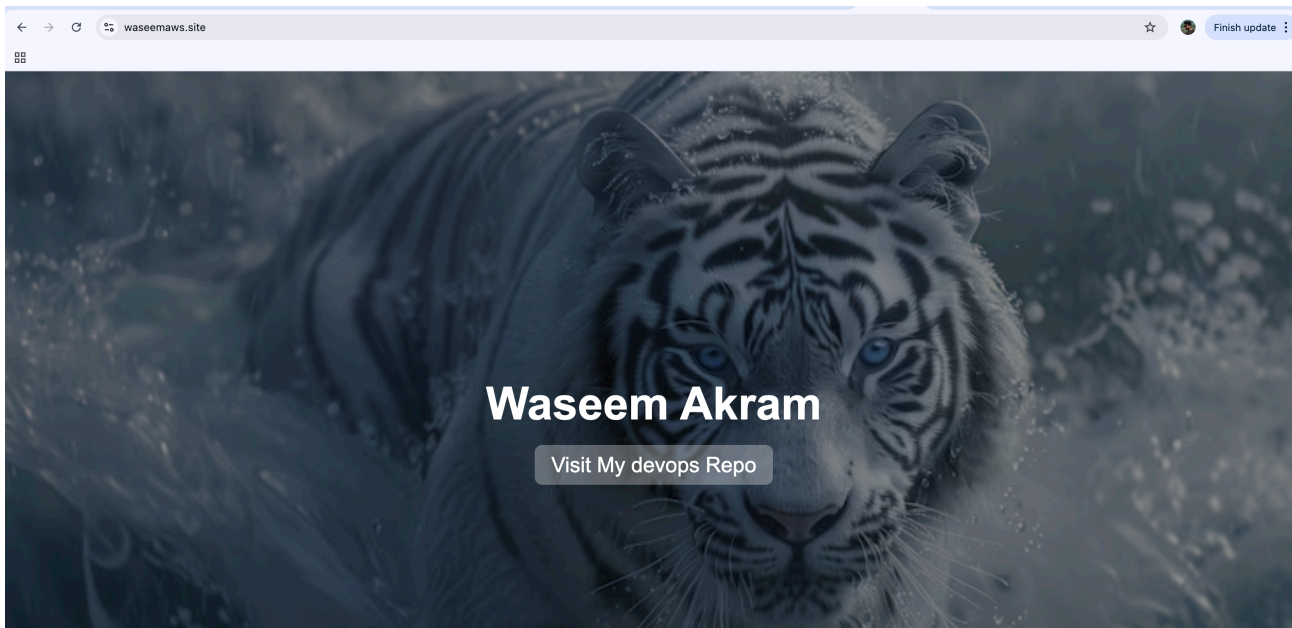
Open

ArgoCD

GitOps CD

Open

—-> refreshing opens the other ec2 web page



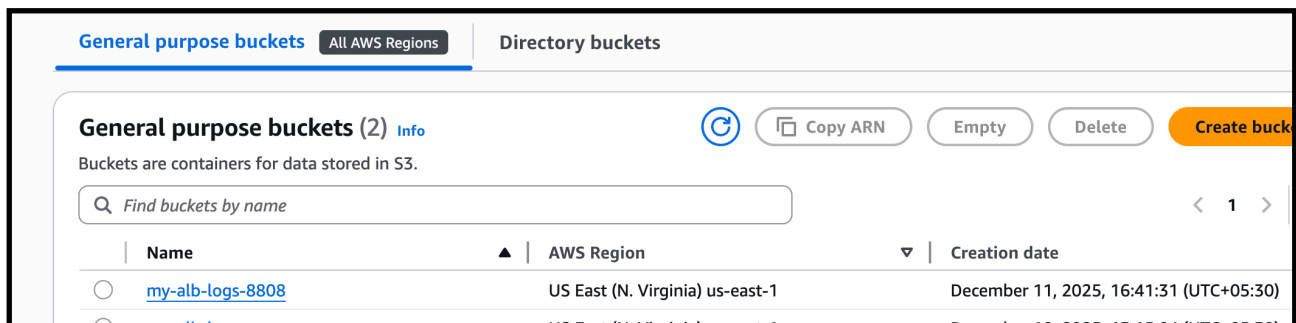
Conclusion

SSL was successfully attached to the Application Load Balancer, enabling secure HTTPS traffic with a valid ACM certificate. The ALB now supports encrypted connections for improved security and compliance.

Task 6:~ Push the application load balancer logs to S3.

STEP 1 — Create an S3 Bucket

1. Go to **S3** → **Create bucket**
2. Bucket name: my-alb-logs-8808
3. **Region must be same as your ALB**
4. Leave everything default → **Create**.



STEP 2 — Add Bucket Policy

Go to **S3** → **Your bucket** → **Permissions** → **Bucket policy** → **Edit**

Bucket policy

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other account

Bucket ARN

 arn:aws:s3::my-alb-logs-8808

Policy

```
1 {  
2   "Version": "2012-10-17",  
3   "Statement": [  
4     {  
5       "Sid": "AWSLogDeliveryAclCheck",  
6       "Effect": "Allow",  
7       "Principal": {  
8         "AWS": "arn:aws:iam::127311923021:root"  
9       },  
10      "Action": "s3:GetBucketAcl",  
11      "Resource": "arn:aws:s3::my-alb-logs-8808"  
12    },  
13    {  
14      "Sid": "AWSLogDeliveryWrite",  
15      "Effect": "Allow",  
16      "Principal": {  
17        "AWS": "arn:aws:iam::127311923021:root"  
18      },  
19      "Action": "s3:PutObject",  
20      "Resource": "arn:aws:s3::my-alb-logs-8808/AWSLogs/207662791773/*",  
21      "Condition": {  
22        "StringEquals": {  
23          "s3:x-amz-acl": "bucket-owner-full-control"  
24        }  
25      }  
26    }  
27  ]  
28 }
```

Save the policy.

STEP 3 — Enable ALB Access Logs

1. Go to **EC2** → **Load Balancers**
2. Select your **Application Load Balancer**
3. Go to **Attributes** tab
4. Click **Edit**
5. Under **Access logs** → Turn **ON**
6. Select your S3 bucket and save

Availability Zone routing configuration

☒ Cross-zone load balancing

Cross-zone load balancing is always on for Application Load Balancers. However, you can turn it off for a specific target group using target group attributes.

ARC zonal shift integration

[Info](#)

Controls whether Amazon Application Recovery Controller (ARC) zonal shift is available to the load balancer.

☒ Disable - Default

Zonal shift will not be available to the load balancer.

☐ Enable

Zonal shift will be available to the load balancer.

Protection

☐ Deletion protection

To prevent your load balancer from being deleted accidentally, turn on deletion protection. If you turn on deletion protection, you must turn it off before you can delete the load balancer.

Monitoring

☒ Access logs

Access logs deliver detailed logs of all requests made to your Elastic Load Balancer. Choose an existing S3 location. If you don't specify a prefix, the logs are stored in the root of the bucket. Additional charges apply. [Learn more](#)

S3 URI

Q

s3://my-alb-logs-8808

X

View

Bro

Format: s3://bucket/prefix

☐ Connection logs

Connection logs deliver detailed logs of all connections made to your Elastic Load Balancer. Choose an existing S3 location. If you don't specify a prefix, the logs are stored in the root of the bucket. Additional charges apply. [Learn more](#)

☐ Health check logs - new

Health check logs deliver detailed logs of health checks for all targets in your Elastic Load Balancer's target groups. Choose an existing S3 location. If you don't specify a prefix, the logs are stored in the root of the bucket. Additional charges apply. [Learn more](#)

Cancel

Save

STEP 4 — Verify

1. Wait 5–10 minutes

2. Go to S3 → Your bucket

Amazon S3 > Buckets > my-alb-logs-8808

my-alb-logs-8808

Objects

Metadata

Properties

Permissions

Metrics

Management

Access Points

Objects (1)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

< 1 >

Name	Type	Last modified	Size	Storage class
AWSLogs/	Folder	-	-	-

Amazon S3 > Buckets > my-alb-logs-8808 > AWSLogs/ > 207662791773/

207662791773/

Objects

Properties

Objects (2)

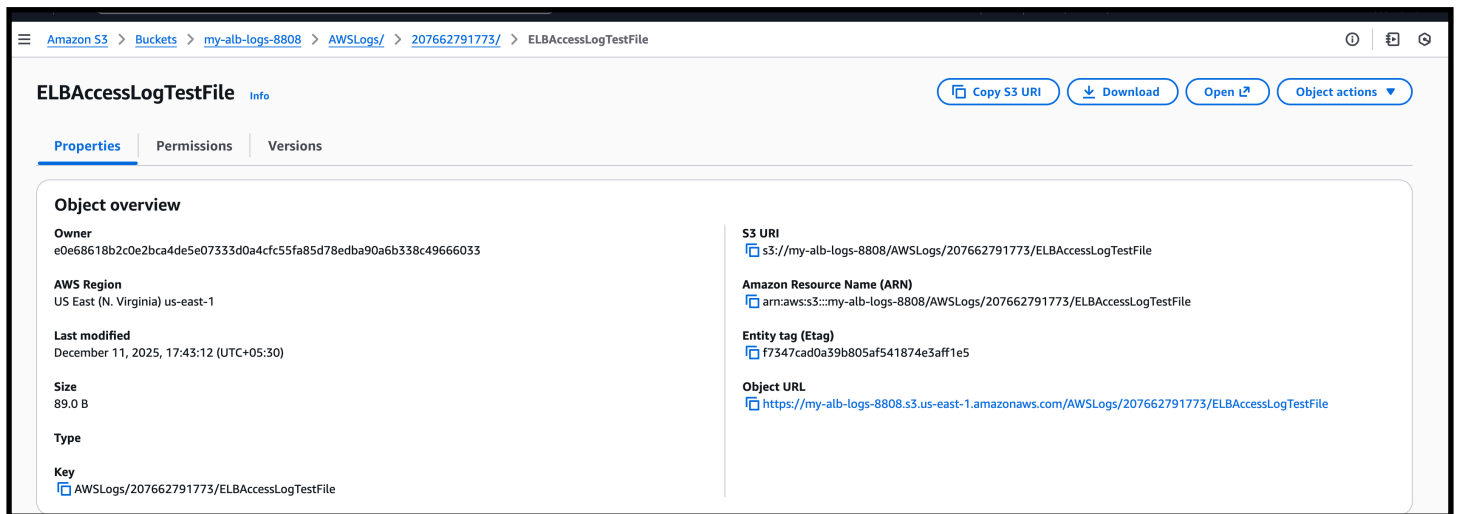
Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

< 1 >

Name	Type	Last modified	Size	Storage class
elasticloadbalancing/	Folder	-	-	-
ELBAccessLogTestFile	-	December 11, 2025, 17:43:12 (UTC+05:30)	89.0 B	Standard

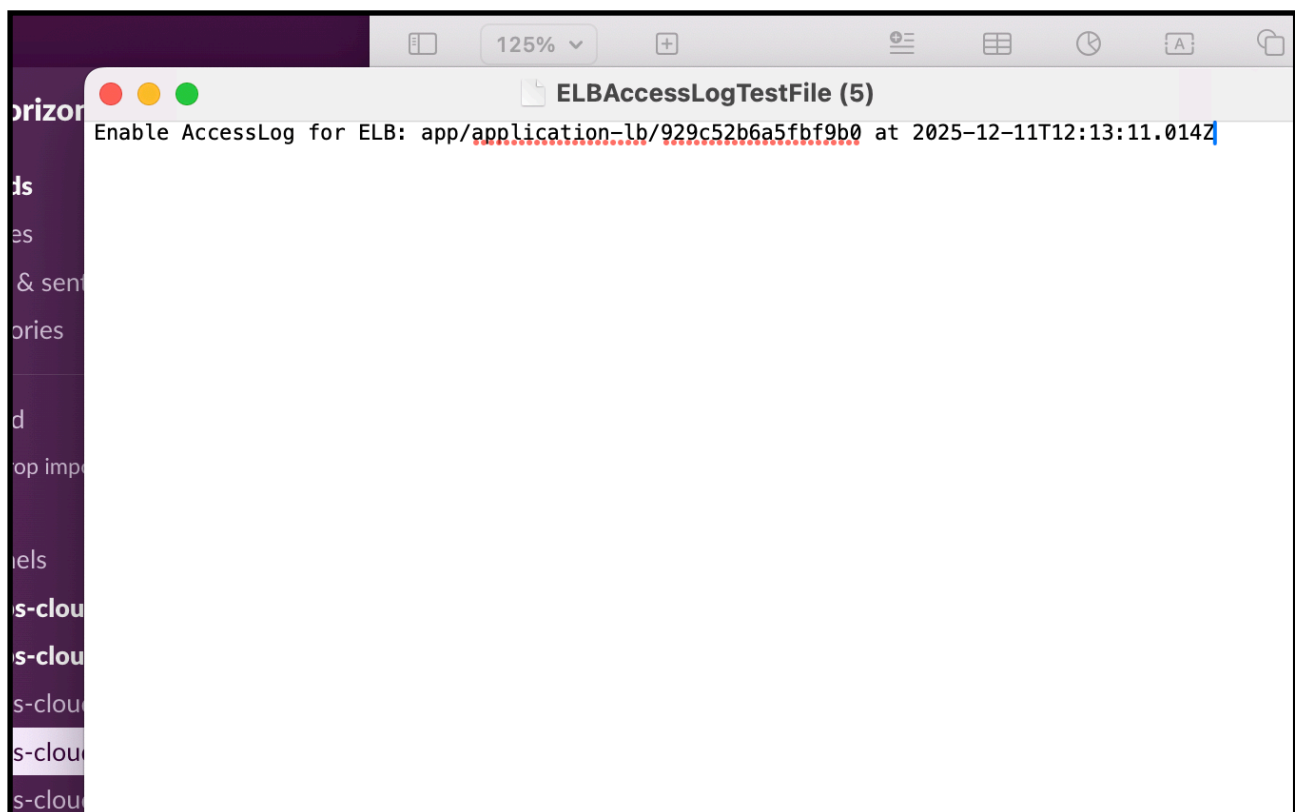
—> Test by Opening Your Website



—>ALB access logs are **NOT webpages**, they are **compressed log files (.gz)**.
So when you click the log file in S3:

✓ **It will download**

✗ **It will NOT open in browser**



Conclusion

ALB access logging was successfully enabled and configured to store logs in the S3 bucket. The logs can now be used for analysis, monitoring, debugging, and compliance reporting.

